

roles	
role_id 	INTEGER NN
role_name	VARCHAR(20) NN
role_level	INTEGER NN
description	VARCHAR(255)

users	
user_id 	INTEGER NN
username	VARCHAR(50) NN
display_name	VARCHAR(100) NN
role_id 	INTEGER NN
is_active	BOOLEAN NN
created_at	DATETIME NN
last_login_at	DATETIME

audit_logs	
audit_id 	INTEGER NN
actor_user_id 	INTEGER
action	VARCHAR(50) NN
target_type	VARCHAR(50) NN
target_id	INTEGER
details_json	JSON
created_at	DATETIME NN

challenges	
challenge_id 	INTEGER NN
title	VARCHAR(100) NN
description	TEXT NN
instructions	TEXT NN
category	VARCHAR(50)
difficulty	VARCHAR(20)
status	VARCHAR(20) NN
time_limit_minutes	INTEGER
created_by_user_id 	INTEGER
created_at	DATETIME NN
updated_at	DATETIME NN

user_credentials	
user_id  	INTEGER NN
password_hash	VARCHAR(255) NN
password_changed_at	DATETIME NN
must_change_password	BOOLEAN NN
failed_login_count	INTEGER NN
locked_until	DATETIME

challenge_instances	
instance_id 	INTEGER NN
user_id 	INTEGER NN
challenge_id 	INTEGER NN
instance_mode	VARCHAR(20) NN
scoring_enabled	BOOLEAN NN
status	VARCHAR(20) NN
requested_at	DATETIME NN
started_at	DATETIME
expires_at	DATETIME
stopped_at	DATETIME
completed_at	DATETIME
network_identifier	VARCHAR(100)
last_activity_at	DATETIME
error_message	TEXT

vm_templates	
template_id 	INTEGER NN
challenge_id 	INTEGER NN
template_name	VARCHAR(100) NN
proxmox_template_vmid	INTEGER NN
proxmox_node	VARCHAR(100) NN
snapshot_name	VARCHAR(100)
vm_role	VARCHAR(50) NN
cpu_cores	INTEGER NN
memory_mb	INTEGER NN
disk_gb	INTEGER
boot_order	INTEGER NN
hostname_prefix	VARCHAR(50)
network_name	VARCHAR(100)
is_user_accessible	BOOLEAN NN
is_active	BOOLEAN NN
created_at	DATETIME NN

vm_instances	
vm_instance_id 	INTEGER NN
instance_id 	INTEGER NN
template_id 	INTEGER NN
proxmox_vmid	INTEGER NN
proxmox_node	VARCHAR(100) NN
hostname	VARCHAR(100)
ip_address	VARCHAR(45)
mac_address	VARCHAR(17)
status	VARCHAR(20) NN
created_at	DATETIME NN
started_at	DATETIME
stopped_at	DATETIME
deleted_at	DATETIME

challenge_flags	
flag_id 	INTEGER NN
template_id 	INTEGER NN
flag_name	VARCHAR(100) NN
flag_hash	VARCHAR(255) NN
points	INTEGER NN
sequence_number	INTEGER
is_active	BOOLEAN NN
created_at	DATETIME NN

instance_jobs	
job_id 	INTEGER NN
instance_id 	INTEGER NN
requested_by_user_id 	INTEGER NN
action	VARCHAR(20) NN
status	VARCHAR(20) NN
attempt_count	INTEGER NN
locked_by	VARCHAR(100)
created_at	DATETIME NN
started_at	DATETIME
completed_at	DATETIME
error_message	TEXT

flag_submissions	
submission_id 	INTEGER NN
user_id 	INTEGER NN
instance_id 	INTEGER NN
vm_instance_id 	INTEGER
matched_flag_id 	INTEGER
was_correct	BOOLEAN NN
was_already_solved	BOOLEAN NN
scoring_enabled	BOOLEAN NN
submitted_at	DATETIME NN

user_solves	
solve_id 	INTEGER NN
user_id 	INTEGER NN
flag_id 	INTEGER NN
instance_id 	INTEGER NN
awarded_points	INTEGER NN
solved_at	DATETIME NN

roles	
role_id 	INTEGER NN
role_name	VARCHAR(20) NN
role_level	INTEGER NN
description	VARCHAR(255)

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username	VARCHAR(50) NN
display_name	VARCHAR(100) NN
role_id 	INTEGER NN
is_active	BOOLEAN NN
created_at	DATETIME NN
last_login_at	DATETIME

audit_logs	
audit_id 	INTEGER NN
actor_user_id 	INTEGER
action	VARCHAR(50) NN
target_type	VARCHAR(50) NN
target_id	INTEGER
details_json	JSON
created_at	DATETIME NN

challenges	
challenge_id 	INTEGER NN
title	VARCHAR(100) NN
description	TEXT NN
instructions	TEXT NN
category	VARCHAR(50)
difficulty	VARCHAR(20)
status	VARCHAR(20) NN
time_limit_minutes	INTEGER
created_by_user_id 	INTEGER
created_at	DATETIME NN
updated_at	DATETIME NN

user_credentials	
user_id 	INTEGER 
password_hash	VARCHAR(255) 
password_changed_at	DATETIME 
must_change_password	BOOLEAN 
failed_login_count	INTEGER 
locked_until	DATETIME

challenge_instances	
instance_id 	INTEGER NN
user_id 	INTEGER NN
challenge_id 	INTEGER NN
instance_mode	VARCHAR(20) NN
scoring_enabled	BOOLEAN NN
status	VARCHAR(20) NN
requested_at	DATETIME NN
started_at	DATETIME
expires_at	DATETIME
stopped_at	DATETIME
completed_at	DATETIME
network_identifier	VARCHAR(100)
last_activity_at	DATETIME
error_message	TEXT

vm_templates	
template_id 	INTEGER NN
challenge_id 	INTEGER NN
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stopped_at	DATETIME
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template_id 	INTEGER NN
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flag_hash	VARCHAR(255) NN
points	INTEGER NN
sequence_number	INTEGER
is_active	BOOLEAN NN
created_at	DATETIME NN

instance_jobs	
job_id 	INTEGER NN
instance_id 	INTEGER NN
requested_by_user_id 	INTEGER NN
action	VARCHAR(20) NN
status	VARCHAR(20) NN
attempt_count	INTEGER NN
locked_by	VARCHAR(100)
created_at	DATETIME NN
started_at	DATETIME
completed_at	DATETIME
error_message	TEXT

flag_submissions	
submission_id 	INTEGER 
user_id 	INTEGER 
instance_id 	INTEGER 
vm_instance_id 	INTEGER
matched_flag_id 	INTEGER
was_correct	BOOLEAN 
was_already_solved	BOOLEAN 
scoring_enabled	BOOLEAN 
submitted_at	DATETIME 

user_solves	
solve_id 	INTEGER NN
user_id 	INTEGER NN
flag_id 	INTEGER NN
instance_id 	INTEGER NN
awarded_points	INTEGER NN
solved_at	DATETIME NN